
0.1 Definitions

The agent is the main entity making the decisions within the environment and responding to feedback from the environment. Each action the agent does nets a reward. High reward might mean a good action in the short run; however, going down that path may not yield a higher total reward. Thus, we introduce the following notion to fix this issue, otherwise the algorithm will act greedily:

Definition 0.1 (Policies/Values). *A policy is a function that takes the current state and decides the next immediate action. A value function predicts the expected total reward after an action is taken. RL models look to improve the value obtained. Policy gradient methods implement only policy functions. Some RL systems implement value functions and implicitly derive the policy function, while some have both as separate processes.*